Presized OTA Simulation

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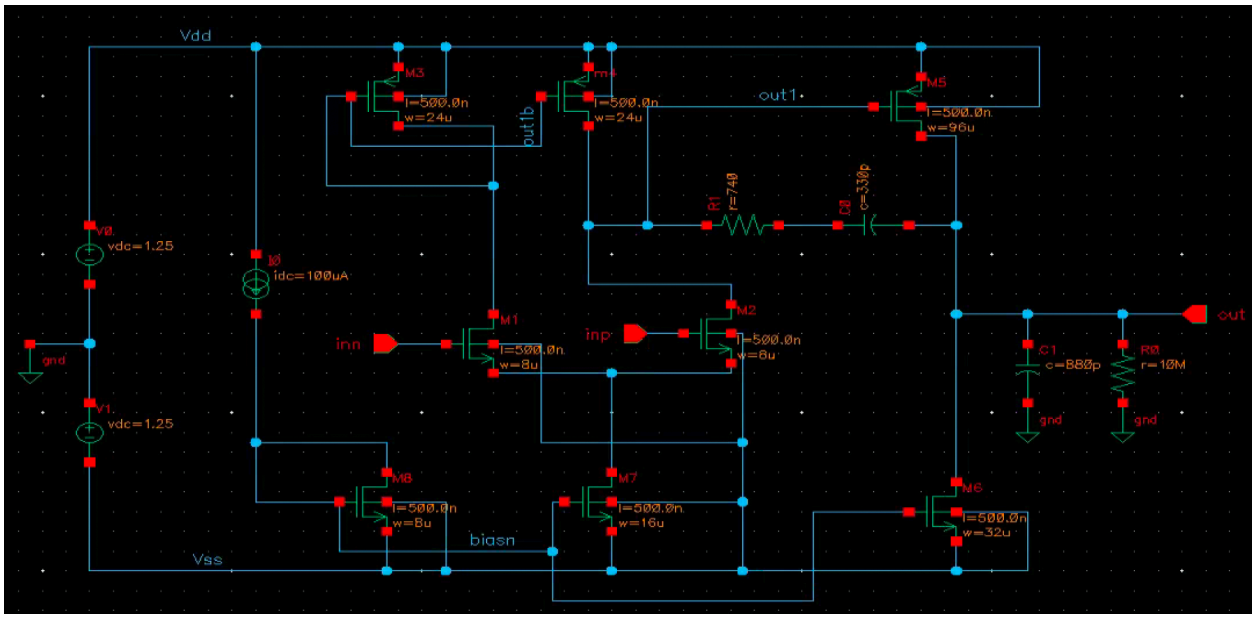
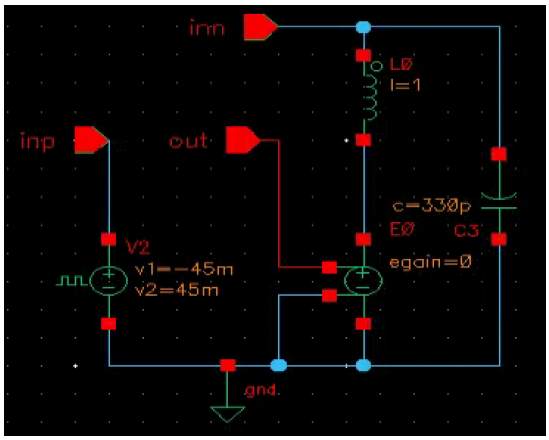


OTA circuit



Closed loop circuit

DC Operating Point Values

Values obtained with closed loop configuration:

Transistor Code

= nmos (diff pair input)	= pmos (diff pair load)	= nmos (current mirror)

Mx (W/L) (um)	VTH (mV)	VDSsat (mV)	VOV (mV)	VDS (mV)	Id (uA)	Gm (mS)	ro (kOhm)	CGG (fF)	Cdd (fF)
M1 (8/0.5)	518.1	217.1	220.6	1200	97.2	0.761	0.015431	23.8	14.2
M2 (8/0.5)	518.1	217.1	220.6	1200	97.2	0.761	0.015431	29.4	32.9
M3 (24/0.5)	-481.4	-264.4	-258.6	-740.0	-97.2	0.632	-0.015431	70.8	50.2
M4 (24/0.5)	-481.4	-264.4	-258.6	-740.0	-97.2	0.632	-0.015431	70.7	44.9
M5 (96/0.5)	-481.4	-264.5	-258.6	-1600	-417.4	2.7	-0.003593	350	407.4
M6 (32/0.5)	414.4	222.1	236.5	905.5	417.5	3.1	0.003592	96	55.2
M7 (16/0.5)	421.0	217.5	229.9	511.3	194.3	1.5	0.007719	52.2	42.2
M8 (8/0.5)	418.7	219.1	232.2	650.9	100	0.755	0.014999	23.9	15.8

*r_o=1/(gamma*Ids)*

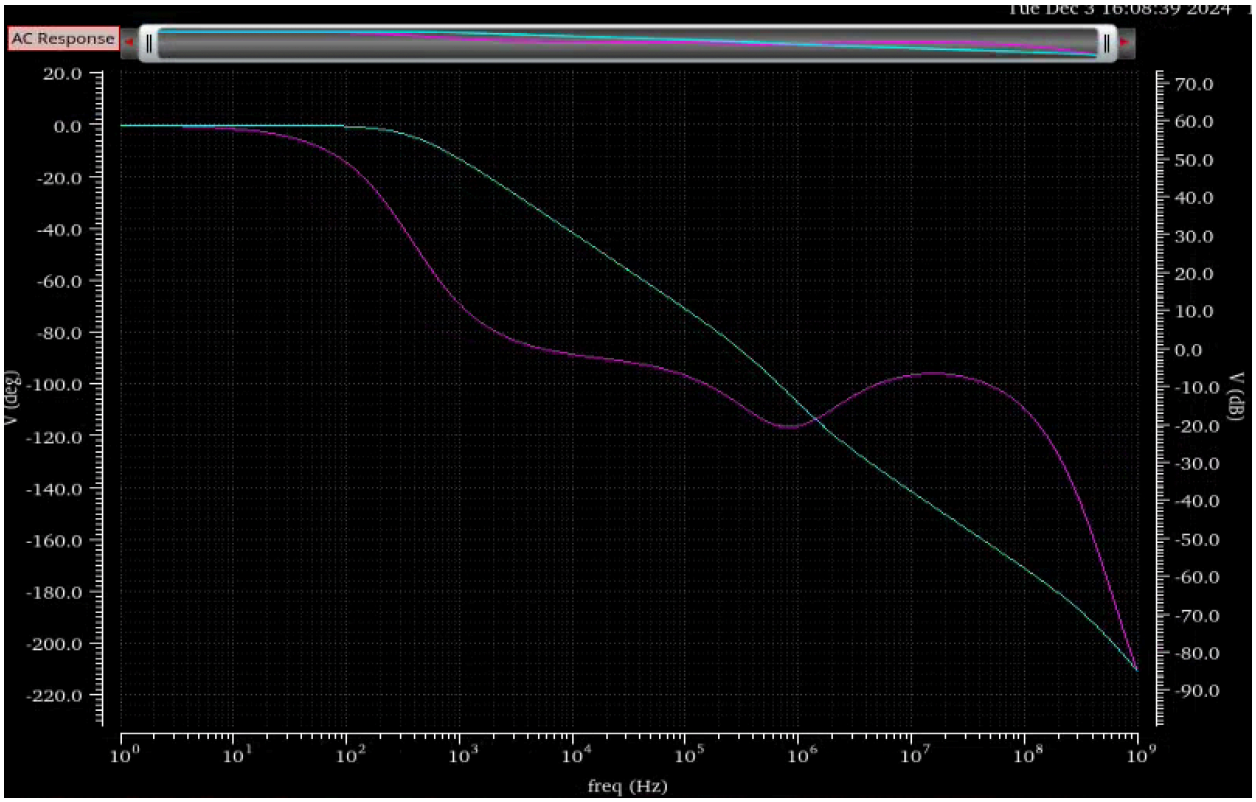
Cgg = Cgs+Cgd

Cdd = Cgd+Cgdovl+Cdb

	M1b	M2b	M3b	M4b	M5b	M6b	M7b	M8b
w	4.0u	4.0u	4.0u	4.0u	4.0u	4.0u	4.0u	4.0u
l	500.0n	500.0n	500.0n	500.0n	500.0n	500.0n	500.0n	500.0n
m	2.0	2.0	6.0	6.0	24.0	8.0	4.0	2.0
as	1.2p	1.2p	1.2p	1.2p	1.2p	1.2p	1.2p	1.2p
ad	1.2p	1.2p	1.2p	1.2p	1.2p	1.2p	1.2p	1.2p
ps	8.6u	8.6u	8.6u	8.6u	8.6u	8.6u	8.6u	8.6u
pd	8.6u	8.6u	8.6u	8.6u	8.6u	8.6u	8.6u	8.6u
ids	97.2u	97.2u	-97.2u	-97.2u	-417.4u	417.5u	194.3u	100.0u
vgs	738.7m	738.7m	-740.0m	-740.0m	-740.0m	650.9m	650.9m	650.9m
vds	1.2	1.2	-740.0m	-740.0m	-1.6	905.5m	511.3m	650.9m
vdsat	217.1m	217.1m	-264.4m	-264.4m	-264.5m	222.1m	217.5m	219.1m
region	saturation	saturation	saturation	saturation	saturation	saturation	saturation	saturation
vbs	-511.3m	-511.3m	0.0	0.0	0.0	0.0	0.0	0.0
vth	518.1m	518.1m	-481.4m	-481.4m	-481.4m	414.4m	421.0m	418.7m
vgsteff	220.5m	220.5m	259.4m	259.4m	259.4m	235.8m	229.1m	231.5m
gm	761.0u	761.0u	632.2u	632.2u	2.7m	3.1m	1.5m	754.8u
gds	14.1u	14.1u	10.4u	10.4u	30.2u	64.3u	45.2u	18.7u
gmb	138.3u	138.3u	205.7u	205.7u	871.3u	711.8u	337.7u	172.9u
gmoverid	7.8	7.8	6.5	6.5	6.4	7.4	7.6	7.5
self_gain	54.1	54.1	60.8	60.8	88.6	48.4	32.6	40.3
cgs	20.3f	20.3f	57.3f	57.3f	229.2f	81.8f	40.9f	20.4f
cgsovl	3.8f	3.8f	13.6f	13.6f	54.3f	15.3f	7.6f	3.8f
cgb	1.1f	1.1f	4.9f	4.9f	19.5f	4.9f	2.5f	1.2f
cgbovl	901.1z	901.1z	2.5a	2.5a	10.0a	3.6a	1.8a	901.1z
cgd	3.5f	3.5f	13.6f	13.6f	54.3f	14.1f	7.1f	3.5f
cgdovl	3.8f	3.8f	13.6f	13.6f	54.3f	15.3f	7.6f	3.8f
cbd	7.0f	7.0f	22.7f	22.7f	77.9f	32.1f	17.6f	51.6a
cjd	7.0f	7.0f	22.7f	22.7f	77.9f	31.9f	17.4f	0.0
cbs	9.9f	9.9f	9.8f	9.8f	39.1f	7.5f	3.8f	1.9f
cjs	8.7f	8.7f	0.0	0.0	0.0	0.0	0.0	0.0
csd	121.4a	121.4a	-17.4a	-17.4a	-2.0a	485.2a	209.1a	115.7a

OTA Characterization Open-Loop Gain

[AC Simulation Bode Plot]



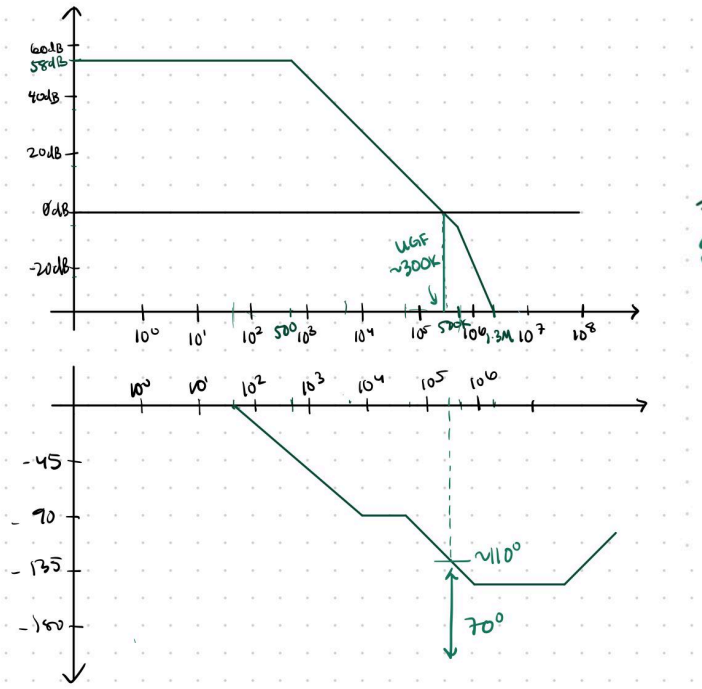
Stability Parameters

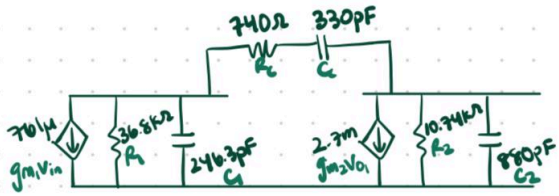
Miller Comp. W/ Series Resistor

($C_c = 330pF, R_c = 740\Omega$)

Parameter	*Calculated Estimate*	*Simulation Value*
DC Gain	812.083V	873
GBW	354032	336978
UGF	300kHz	298.102kHz
PM	70°	72.4°
f_{p1}	436.87 Hz	386 Hz
f_{p2}	487.8 kHz	508 kHz
f_z	1.304 MHz	1.32 MHz

Calculated Estimate Work:





$$f_1 = \frac{1}{2\pi} \frac{1}{R_1(C_1 + (1+g_{m_2}R_2)C_2)} = 436.87\ Hz$$

$$f_2 = \frac{1}{2\pi} \frac{g_{m_2}C_c}{C_1C_2 + C_1C_c + C_2C_c} = 487.8\ kHz$$

$$f_3 = \frac{g_{m_2}}{(1-g_{m_2}R_c)C_c} = 1.304\ MHz$$

Simulated Estimate:

	Real	Imaginary	Qfactor
1	-3.86045e+02	0.00000e+00	5.00000e-01
2	-5.07572e+05	0.00000e+00	5.00000e-01
3	-5.34922e+08	0.00000e+00	5.00000e-01
4	-6.74954e+08	0.00000e+00	5.00000e-01
5	-8.13756e+08	0.00000e+00	5.00000e-01
6	-4.65976e+09	0.00000e+00	5.00000e-01

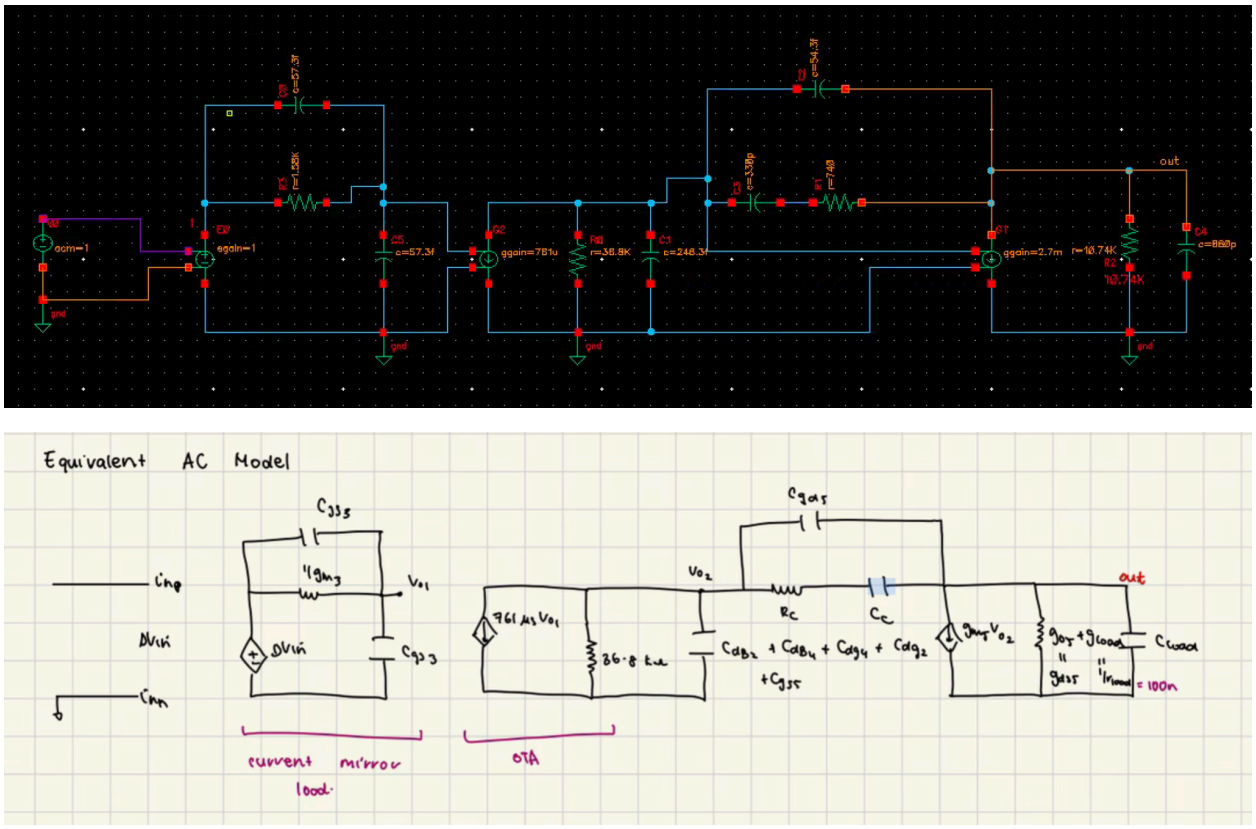
Zeros (Hz)
at V(out,inn)/V8

	Real	Imaginary	Qfactor
1	-1.31575e+06	0.00000e+00	5.00000e-01
2	-7.05657e+08	0.00000e+00	5.00000e-01
3	7.14771e+08	0.00000e+00	-5.00000e-01
4	-1.41955e+09	+/- 6.66889e+08	5.52427e-01
5	-2.50725e+10	0.00000e+00	5.00000e-01

Constant factor = 5.72450e-06

DC gain = 8.73419e+02

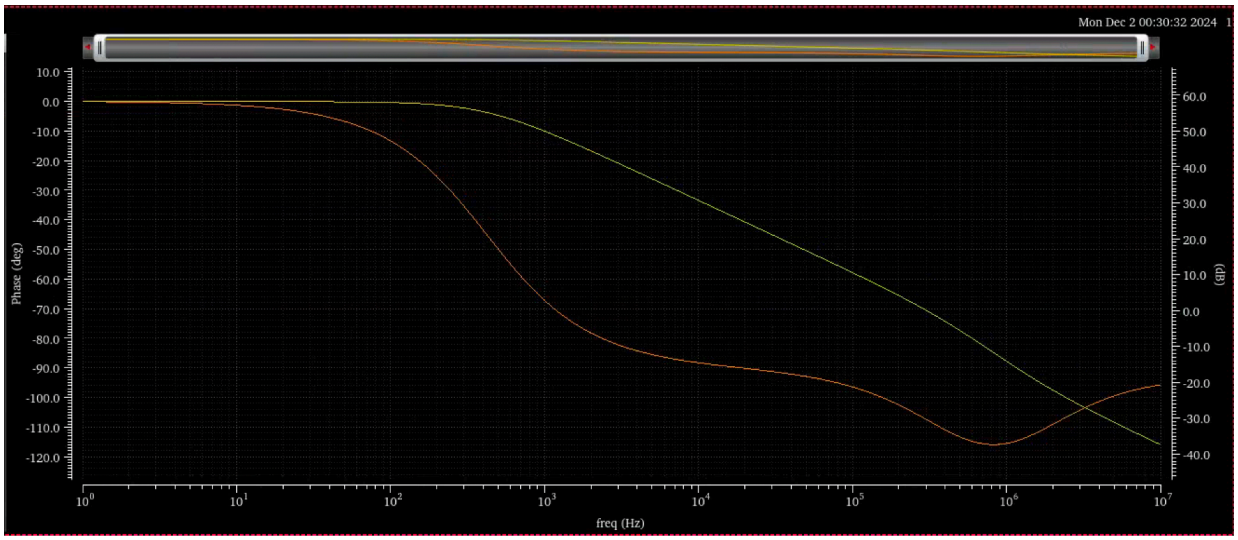
Small Signal Equivalent Open Loop



[SS Eq. Circuit Schematic]

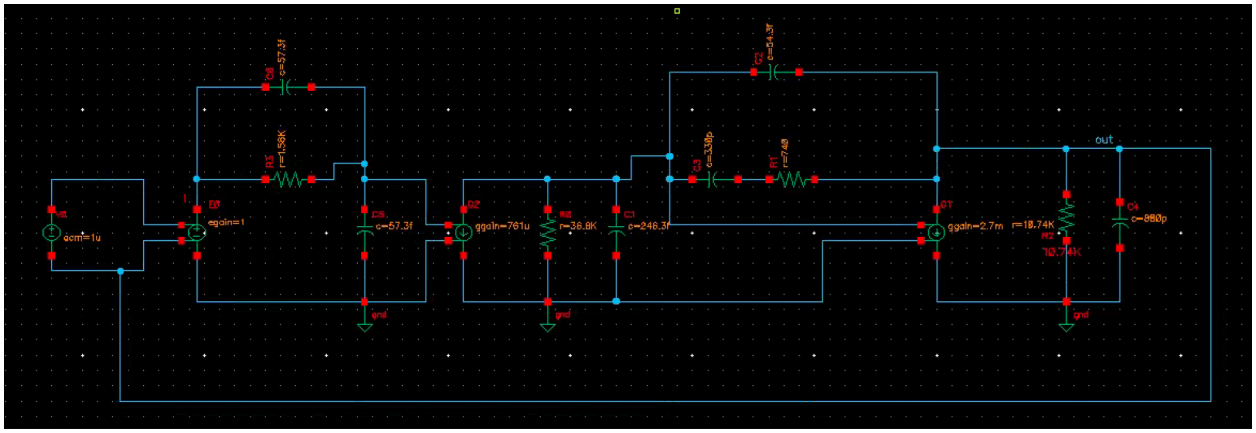
Parameter	*SS Eq. Circuit Simulation Value*
DC Gain	812
GBW	$812*4.21 * 10^2 = 348689.04$
UGF	302kHz
PM	72.5°
f_{p1}	$4.21 * 10^2$
f_{p2}	$5.13 * 10^5$
f_z	$1.3 * 10^6$

```
****************
PZ Analysis `pz'
****************
DC simulation time: CPU = 110 us, elapsed = 110.865 us.
DC simulation time: CPU = 40 us, elapsed = 38.8622 us.
                    Poles (Hz)

          Real                      Imaginary                 Qfactor

   1   -4.21832e+02                0.00000e+00              5.00000e-01
   2   -5.12555e+05                0.00000e+00              5.00000e-01
   3   -7.30279e+08                0.00000e+00              5.00000e-01
   4   -8.78979e+08                0.00000e+00              5.00000e-01

                    Zeros (Hz)
                    at V(out,0)/V0

          Real                      Imaginary                 Qfactor

   1   -1.30457e+06                0.00000e+00              5.00000e-01
   2   -1.75796e+09                0.00000e+00              5.00000e-01
   3    3.95358e+09                0.00000e+00             -5.00000e-01

Constant factor =  7.81018e+04

DC gain =  8.12083e+02
```

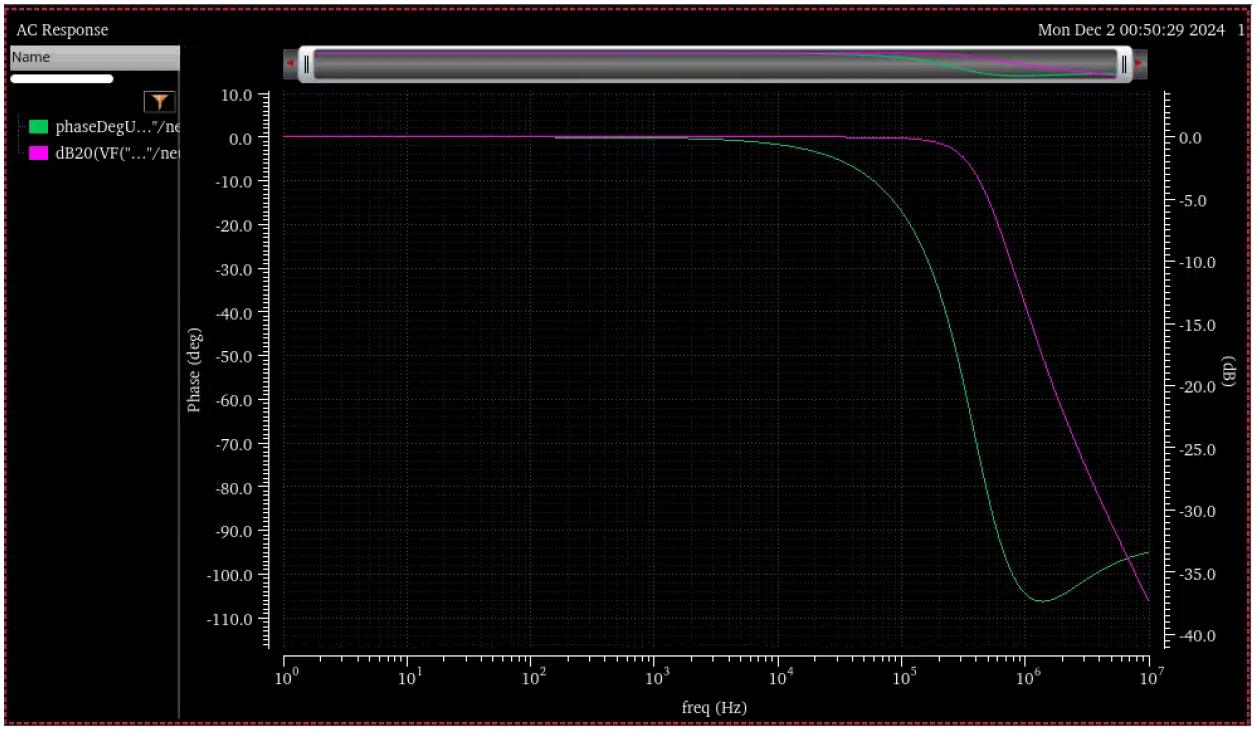


[AC Simulation of SS Eq. Circuit Bode Plot]

Small Signal Equivalent Closed Loop



[SS Eq. Circuit Schematic]



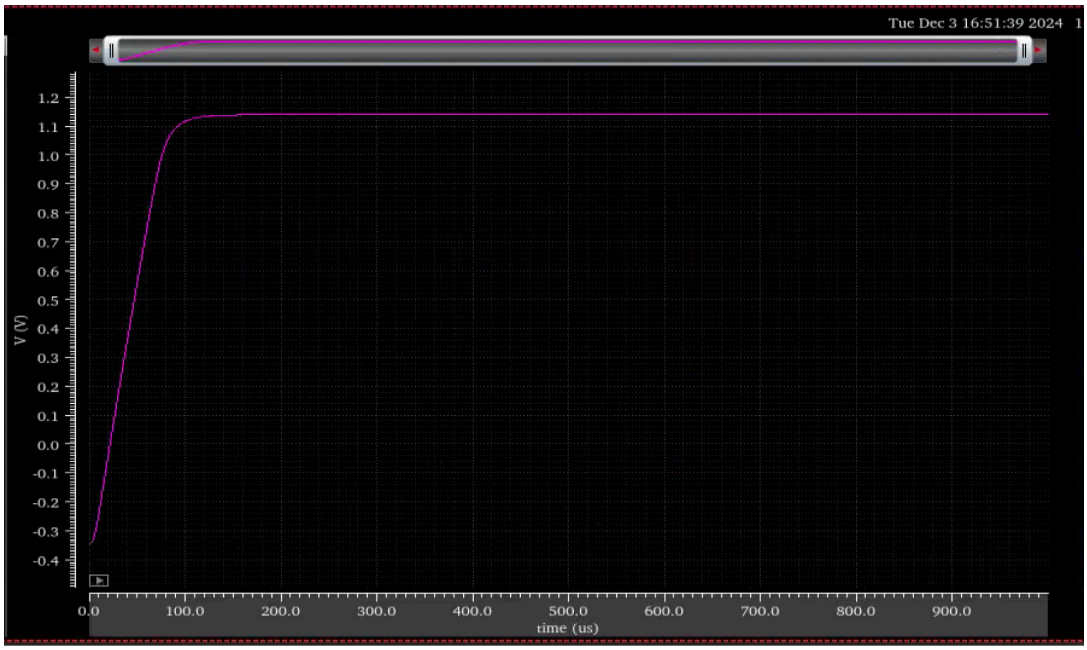
[AC Simulation of SS Eq. Circuit Bode Plot]

```
****************
PZ Analysis `pz'
****************
DC simulation time: CPU = 149 us, elapsed = 150.919 us.
DC simulation time: CPU = 84 us, elapsed = 82.9697 us.
                      Poles (Hz)

           Real                        Imaginary                Qfactor

   1   -4.21832e+02                 0.00000e+00              5.00000e-01
   2   -5.12555e+05                 0.00000e+00              5.00000e-01
   3   -7.30279e+08                 0.00000e+00              5.00000e-01
   4   -8.78979e+08                 0.00000e+00              5.00000e-01

                      Zeros (Hz)
                      at V(out,0)/V0

           Real                        Imaginary                Qfactor

   1   -1.30457e+06                 0.00000e+00              5.00000e-01
   2   -1.75796e+09                 0.00000e+00              5.00000e-01
   3    3.95358e+09                 0.00000e+00             -5.00000e-01

Constant factor =  7.81018e+04

DC gain =  8.12083e+02
```

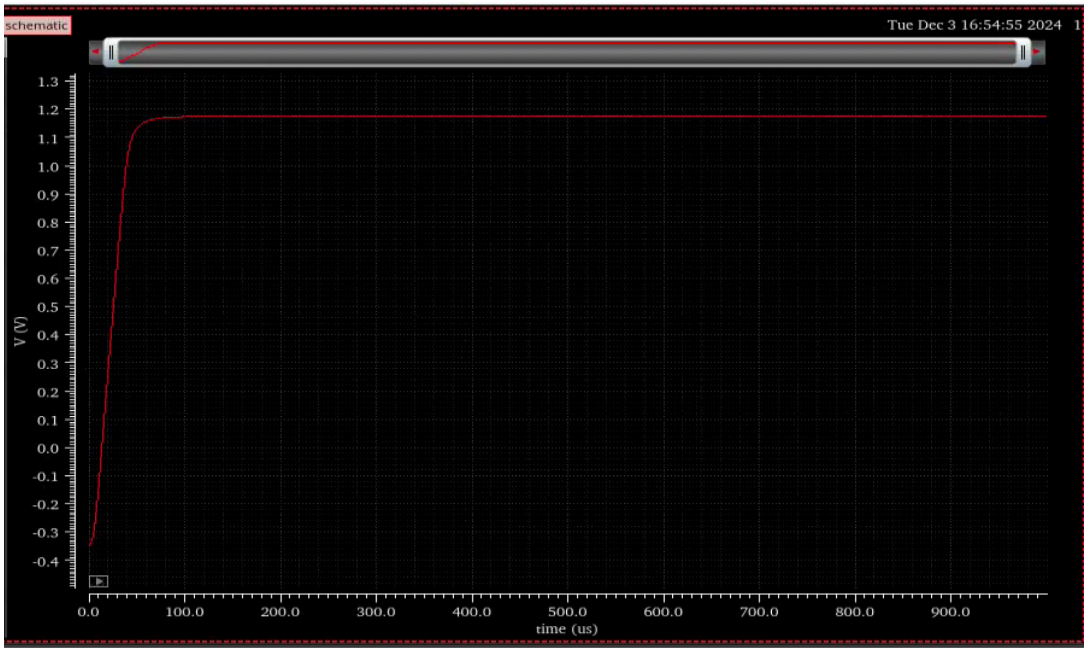
[PZ Values]

Unity gain frequency of our open loop system should be the gain bandwidth of our closed loop system. We are not getting that

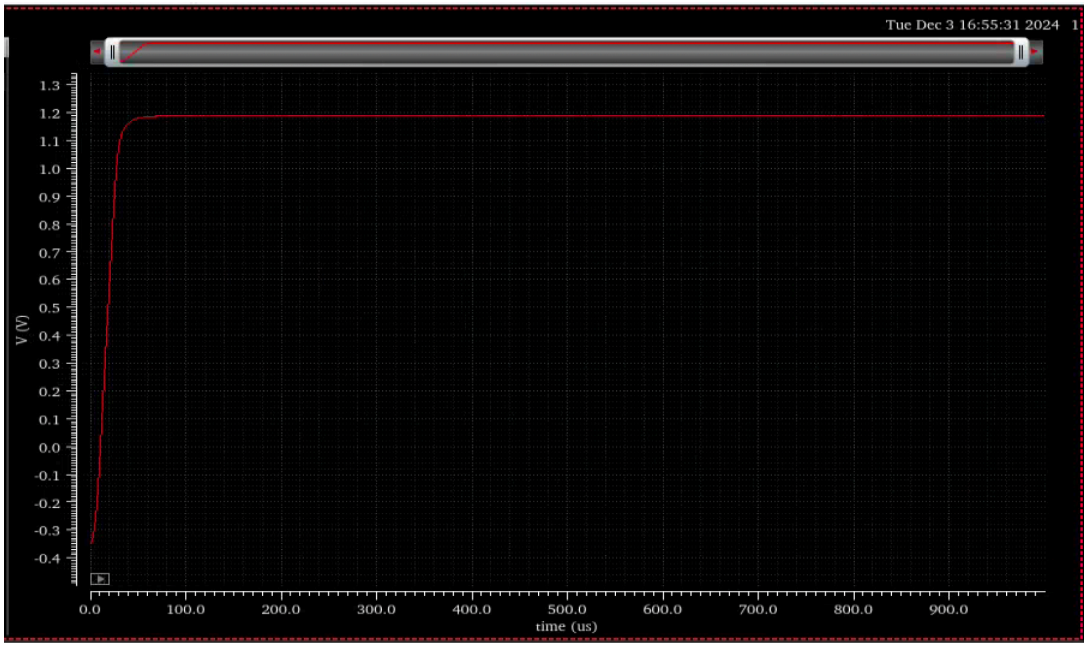
Step Response for the closed-loop configuration



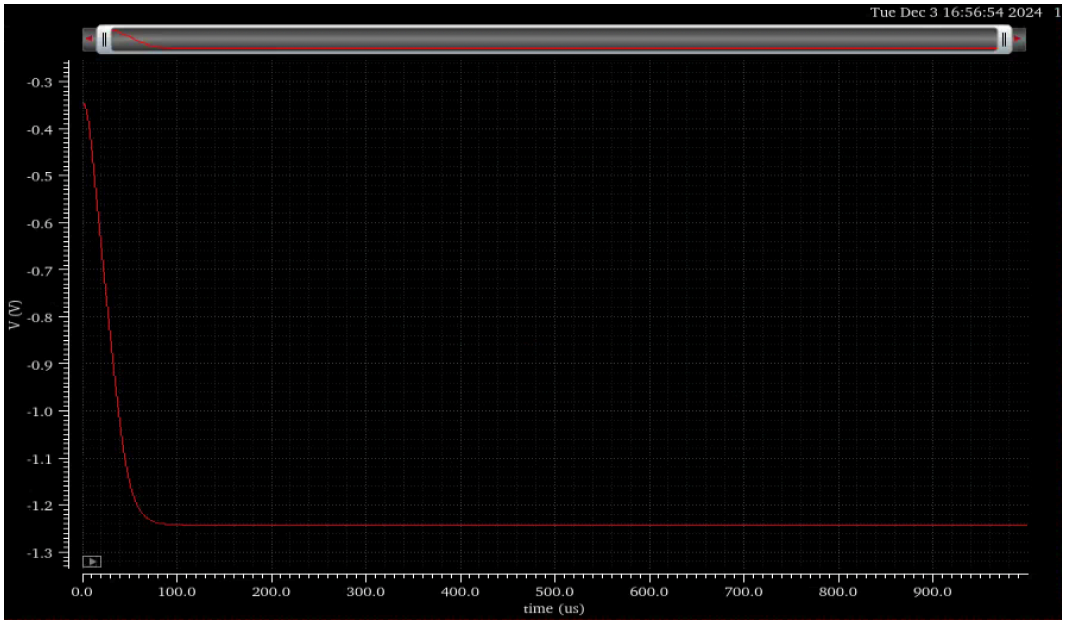
10mV small step



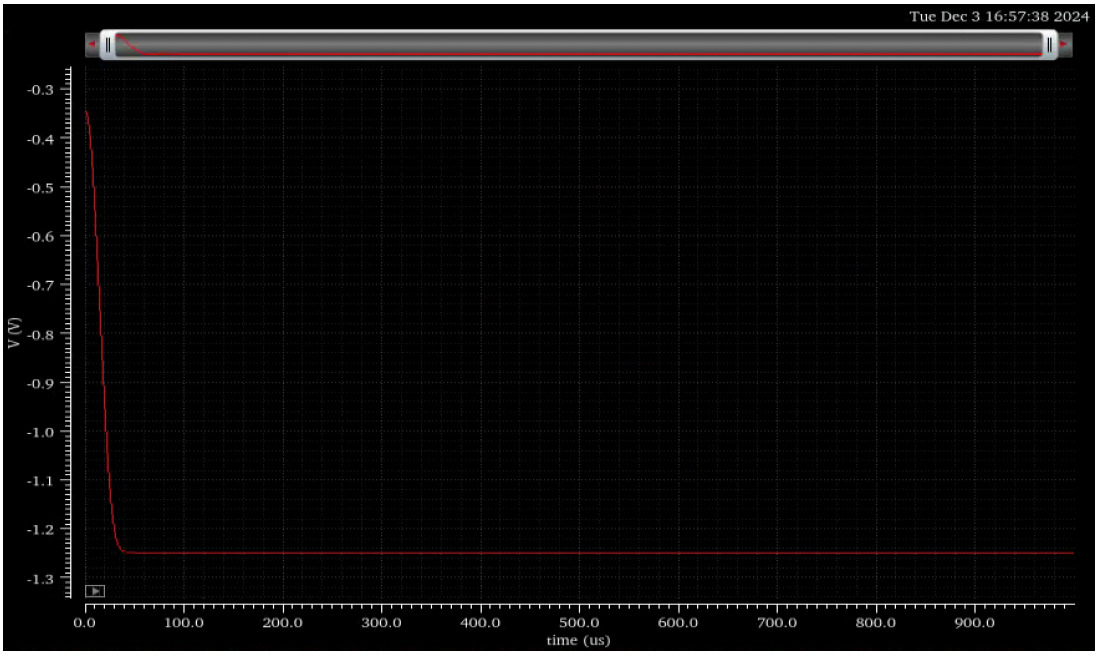
20mV small step



30mV small step



-10mV small step



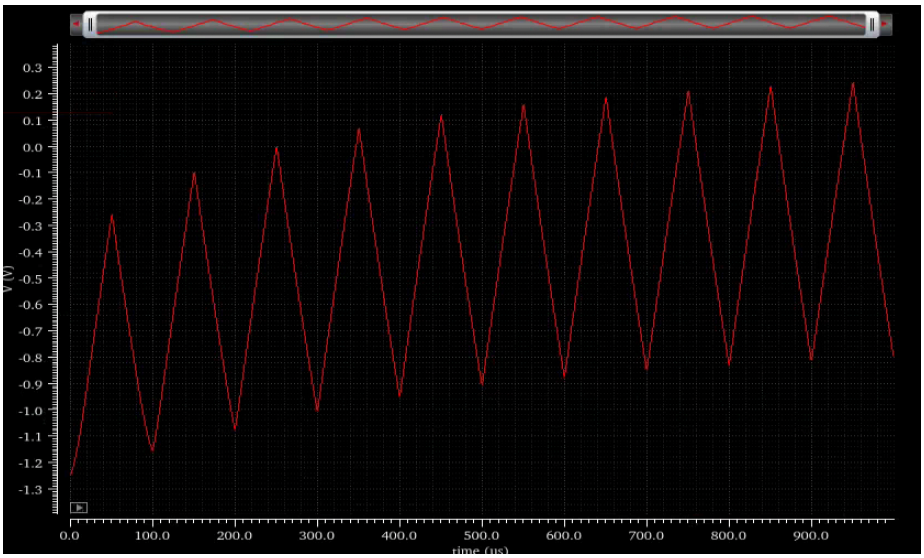
-20mV small step

As the step input is increased, the output current increases proportionally. This can be observed by the steepness of the linear region increasing as the input increases in voltage. Similarly, when applying negative step inputs, the output gets steeper with more negative values.

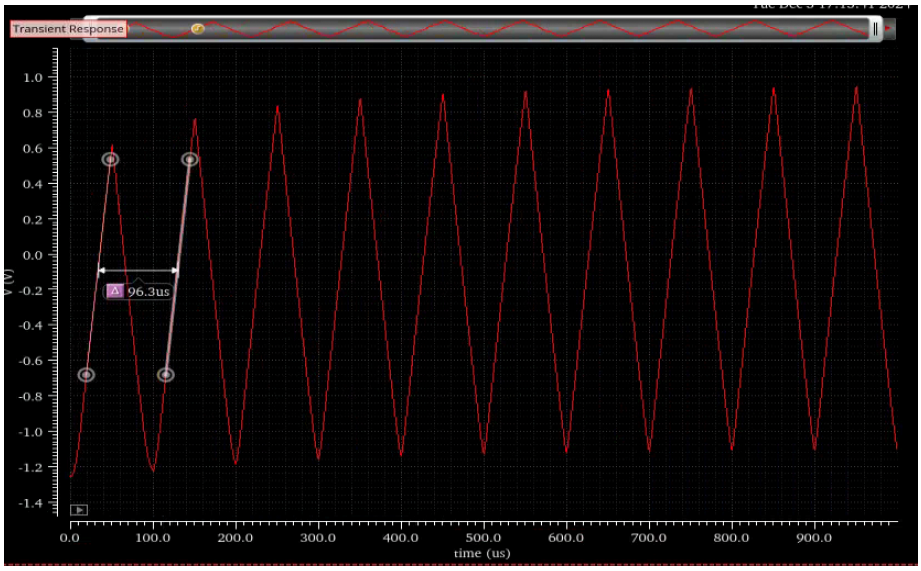
This behavior is expected for an OTA operating in the linear region, where the output is directly proportional to the input voltage and the g_m.

Pulse Train Response for the closed-loop configuration

For a 10kHz frequency, we used 100us period and a 50u pulse width

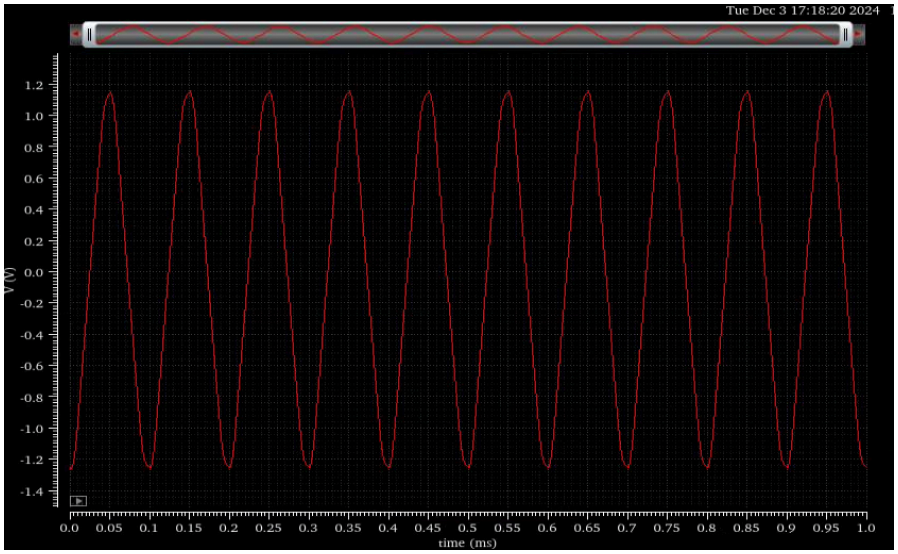


10mV step input



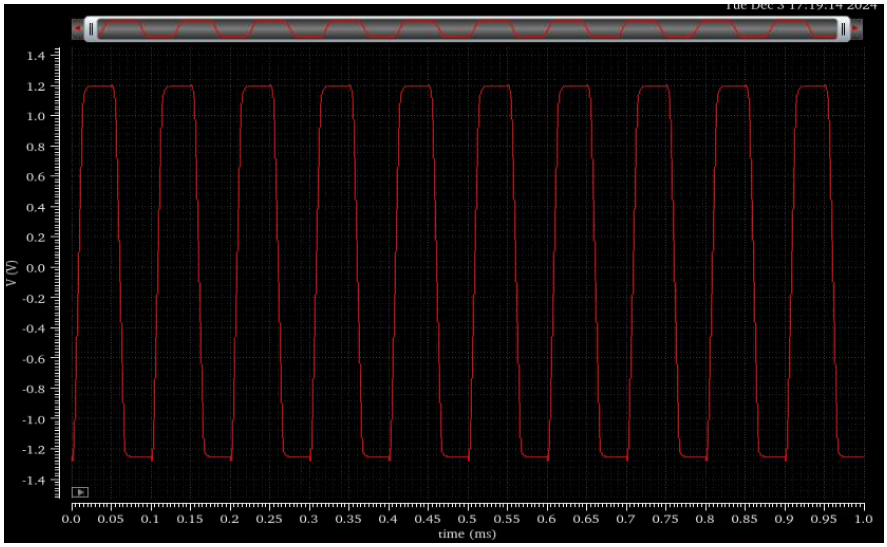
20mV step input

Verifying frequency: $f = 1/T$ = 1/96.3us = 10.4kHz (correct)

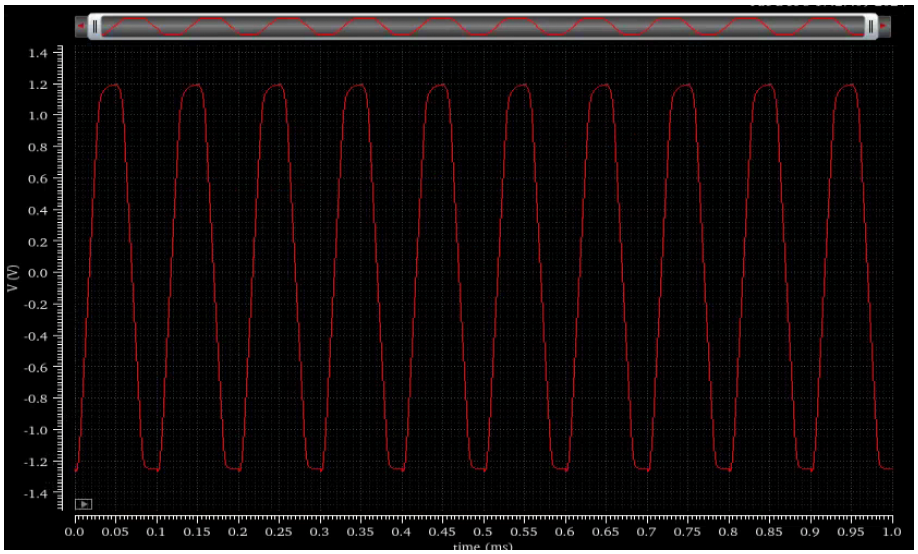


30mV step input

The pulse train changes linearly with the input step amplitude. The output waveform scales according to g_m, as shown in the slope of the waveform increasing proportionally.



100mV step input



45mV step input

Seem to get significant clipping after around 45mV step input. When increased input to 100mV (10x larger), got distorted graph with clipping, too high of frequency of the circuit to handle.